Track A — Air Vehicle

RescueSwarm: work package and funding request

NIDAR 2026–27 · Track 1 · Mission 1

Ask: INR 334,022 · 2 students · Frame, propulsion, power, payload mechanism, assembly

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This track builds the thing that flies. Everything else in the programme is carried by it, which means this track's schedule is the programme's schedule: no other track can begin flight validation until an aircraft exists, and the single measurement that decides whether one does is owned here.

1 Scope and ownership

This track owns **frame, propulsion, power, payload mechanism, assembly**. It is one of four delivery tracks defined in `docs/development-plan.md` §1.3; the integrated system argument lives in the master proposal, `docs/proposal/rescueswarm-proposal.pdf`, which this document does not restate.

2 The design point this track is funded to build

The aircraft is a quadrotor on 18 in propellers with a maximum take-off mass of **6.36 kg**, giving a three-aircraft fleet of **19.08 kg** against the 25 kg regulatory cap — 24 % margin. Structure is 1 495 g (23.5 % of MTOW) and the battery pack 1 449 g (22.8 %); together they are 46 % of the aircraft, which is why airframe mass fraction and pack chemistry are the two sensitivities this track carries in the risk register.

All of these figures regenerate from `tools/sizing-model/rescueswarm_sizing_model.py`. None is transcribed by hand.

3 Propulsion, and the measurement that is load-bearing

At a thrust-to-weight ratio of 2.0 the aircraft requires **125 N total**, which is **3.18 kgf per motor** on an 18 in propeller, with hover sitting at 1.59 kgf — 50 % of maximum, inside the healthy 45–55 % band. The mass budget allows 160 g per motor.

The funded motors publish no thrust curve. That is a deliberate cost decision, not an oversight: professional-grade motors with published data cost roughly 2.4 times as much. The consequence is that the thrust-to-weight figure above is a *requirement* rather than a measurement until the thrust stand runs in P2, and the thrust stand is therefore not a convenience — it is the instrument that says whether this aircraft hovers. It is funded accordingly.

Two figures in `docs/sizing/sizing-calculations.md` contradict the model here, quoting 2.53 and 2.94 kgf per motor against the model's 3.18, and one line still specifies a 20 in propeller where the bill of materials buys 18 in. Reconciling those is a P1 action for this track.

4 Energy: what the pack must actually do

The pack is sized by a reserve policy, not by the design mission. The mission consumes 105.5 Wh; the policy requires the pack to also carry one complete re-sweep and four minutes of loiter, all inside an 80 % depth of discharge:

$$105.5 + 22.1 + 60.8 = 188.4 \text{ Wh usable,}$$

against 233.3 Wh available — a **24 % margin**.

Stated independently of chemistry, so that any candidate pack can be tested against it, the requirement is **6S**, **≥236 Wh** nameplate, **≥115 A continuous**, **≤40 mΩ** and **≤1 450 g**. The current figure is the one most often missed: it is not a burst rating. At T/W 2.0 the aircraft draws 2 481 W, which at 21.6 V is 115 A, and that must be available continuously.

The adopted implementation is 18 cells in 6S3P at 4500 mAh and 45 A continuous. A 40 A cell leaves 4.5 % burst margin against 18 %, and cells whose 45 A rating depends on an 80 °C cut-off are 35 A cells in Indian ambient — below the 38.3 A per-cell peak. The specification was tightened accordingly on 2026-08-20.

5 Two open items this track owns

The mass statement does not close. Itemised masses sum to 6 061 g against a 6 360 g MTOW, leaving **299 g** — 4.7 % — unattributed. It is small enough not to threaten fleet margin and large enough that it should be attributed rather than absorbed. Closing it is a P1 action.

Cell sourcing is a live commercial question. The budget carries INR 700 per cell against a domestic supplier quote. A drop-in equivalent lists publicly at INR 405. The action is not to switch — cells are one of only four remaining Indian lines and indigenisation is scored — but to hold the supplier quote against a published benchmark before committing.

6 Funding request

Every figure below is generated from `docs/proposal/figures/track_budget.py`, which partitions the master competition budget. The four track asks plus the shared pool reconcile to the master figure exactly; that reconciliation is asserted in code and fails the build if it ever stops holding. **K** marks a line held at professional grade on purpose.

Item		INR
Motors (4 x 4,500 x 3 ac)		54,000
Structure (1 x 13,000 x 3 ac)	**K**	39,000
Li-ion cells (18 x 700 x 3 ac)	**K**	37,800
Pack, BMS, PDB, BEC (1 x 8,500 x 3 ac)		25,500
ESCs (4 x 1,800 x 3 ac)		21,600
Propellers (4 x 1,625 x 3 ac)		19,500
Battery chargers		14,000
Payload system (1 x 4,500 x 3 ac)		13,500
Calibrated scale + remaining		12,000
Thrust stand		8,000
Wiring, connectors (1 x 2,400 x 3 ac)		7,200
Prop adapters (4 x 350 x 3 ac)		4,200
Subtotal		256,300
Customs duty and freight		13,949
GST		20,205
Contingency at 15%		43,568
TRACK A ASK		**334,022**

Deferred or institutionally held, and therefore not requested: 3D printer, Cell tester, Charger PSU, Recovery parachutes, 3, Relief kits (14), Rotary tool + CF extraction, Spare airframe structure set, Spare battery packs, Spare motors, Spare propellers, plate and tube stock. These remain capabilities the track requires; they are simply not being bought. Where a deferral carries a technical consequence it is stated in the risk table below rather than left implicit.

7 Deliverables by phase

Phase	Dates	This track delivers
P1	24 Aug – 13 Sep	Long-lead orders placed: cells, motors, structure stock. Mass residual attributed. Sizing-document contradictions reconciled.
P2	14 Sep – 4 Oct	**Thrust-stand measurement** – the gate for the whole programme. Pack assembled and bench-discharged.
P3	5 Oct – 18 Oct	First flight. Measured mass and hover current reported to Track C.
P5	9 Nov – 22 Nov	Single-aircraft full mission including payload release.

8 Risks owned by this track

Risk	Sev.	Mitigation
Motors deliver less than 3.18 kgf	High	P2 thrust stand. If they miss, the fallback is a published-curve motor at roughly 2.4× the cost, which is a budget event this track must flag the moment it is known.
Mass residual grows during build	Medium	299 g is unattributed today. Fleet carries 4 919 g of growth allowance, so this is affordable, but it must be tracked from P1 not discovered at weigh-in.
Cell lead time	Medium	No spare packs are funded. Domestic sourcing is 2–3 weeks; order at P1, not P2.

9 Interfaces to other tracks

With	Dependency
Track B	Provides the power distribution the avionics run on, and the vibration-isolated mounting the flight controller requires.
Track C	Provides the mass, thrust and endurance figures the SITL model flies against. If the built aircraft differs from the model, the simulations stop being evidence.
Track D	Provides the camera mounting plane and its boresight stability. Geolocation accuracy depends on that mount not moving.

Interfaces are where student programmes fail, because each side assumes the other owns the gap. Every row above is a two-way commitment and should be recorded as an interface control document at P1.

Generated by `tools/proposal/build_track_proposals.py` from `docs/proposal/figures/track_budget.py`. Financial figures are not maintained in this document and must not be edited here: change the master budget and regenerate. Technical claims cite the model or document that produced them.